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TDEM GEOPHYSICAL SURVEY AT THE NAHAL HEVER NORTH SITE

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by

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INTRODUCTION

The Geophysical Institute of Israel (GII) carried out the geophysical survey at the Nahal Hever North (NHN) site using the time domain electromagnetic (TDEM) method. The survey is the continuation of the similar survey carried out in 1998 at the Nahal Hever South (NHS) site (Shtivelman et al., 1999). The current survey included six measurements along the NW-SE profile (Fig. 1). The measurements were performed using adjacent 50x50 m central loop arrays. The location map in Fig. 1 shows the location of the receiver coils only.

Exactly as in the previous survey, the measurements at the NHN site were carried out using both EM-47 and EM-67 TDEM systems and the data were processed and interpreted using the TEMIX-XL interpretation package (for further details see Shtivelman et al., 1999).

RESULTS

The results of the interpretation are presented in Figures 2(a,b) and 3(a,b) in the form of pseudo-2D resistivity cross-sections based on the layered and smooth inversion algorithms, respectively (Shtivelman et al., 1999). Figures a) and b) differ only in the colour scales used: the colour scales used in Figures 2a and 3a are identical to those ones used in the NHS cross-sections (Shtivelman et al., 1999). The colour scales used in Figures 2b and 3b significantly narrow the range of the very low resistivity unit. Similarly to the NHS cros-sections (Shtivelman et al., 1999) the interpreted resistivities can be divided in roughly four groups represented by different colours in Figures 2 and 3:

- Very low resistivities (less than 1 ohm-m in Figures 2a and 3a and less than 0.3 ohm-m in Figures 2b and 3b) shown in red. This resistivity unit is identified with the Dead Sea water saturated sediments (both aquifers and aquicludes).
- Low resistivities (between 1 and 2 ohm-m in Figures 2a and 3a and between 0.3 and 2 ohm-m in Figures 2b and 3b) shown in pink. This unit represents saline water saturated aquifers and aquicludes when the water salinity varies between the Dead Sea and normal sea water salinities.
- Moderate resistivities (between 2 and 4 ohm-m) shown in light blue. Typical for brackish water saturated aquifers and some brackish to fresh water saturated and unsaturated aquicludes.

- High resistivities (greater than 4 ohm-m) shown in blue to dark blue. These resistivities (especially those shown in dark blue) are typical for either dry or fresh-water saturated aquifers.

DISCUSSION

Comparison of the layered and smooth interpretation results (Figures 2a,b and 3a,b respectively) shows that practically no difference exists between them, if the very low resistivity unit is defined within the range between 0 to 1 ohm-m (compare Figures 2a and 3a). Both cross-sections are represented by approximately 18 m thick high resistivity layer (dry or fresh water saturated aquifer) underlain by a very thin low resistivity unit, most likely representing transistion zone, followed by the Dead Sea water intrusion going down to the maximum penetration depth of up to 100 m. The fresh/sea water interface is practically horizontal due to the direction of the profile, which is nearly parallel to the coastal line. Contrary to the NHS cross-sections (Shtivelman et al., 1999), no resistivity anomalies are detected within the Dead Sea water intrusion.

For the narrower very low resistivity range, varying between 0 and 0.3 ohm-m, the layered and smooth interpretations are slightly different (compare Figures 2b and 3b). The main difference manifests itself in somewhat wider transition zone and in the existence of two local resistivity anomalies within the very low resistivity unit according to the smooth inversion. Since these features are detected by only one of the inversions, we consider them as questionable (Shtivelman et al., 1999). Another feature, which was detected by both inversions along the whole profile, can be considered as very reliable. This feature is represented by the appearance of a low resistivity layer below the very low resistivity one. The depth to the top of the low resistivity layer varies along the profile, namely it becomes significantly shallower below the NW end of the profile. Since the resistivity of this layer varies in the narrow range between 0.3 to 0.5 ohm-m, the hydrogeological interpretation of this resistivity feature is very questionable. On the one hand, it can be identified with the salinities slightly lower than those of the Dead Sea water (in the range of 100 to 200 g/l). On the other hand, it may reflect lithological changes leading to much lower porosities, while the water salinity can remain the same. The sparse calibration measurements available for the Dead Sea coastal aquifer (Kafri et al., 1997) do not allow to resolve the above

non-uniqueness, although the explanation based on the water salinity change has slightly greater statistical significance.

REFERENCES

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Shtivelman, V., Goldman, M., Ronen, A. and Ezersky, M., 1999. Shallow geophysical surveys in the sinkhole development area at the Nahal Hever site, GII report No. 823/87/98.

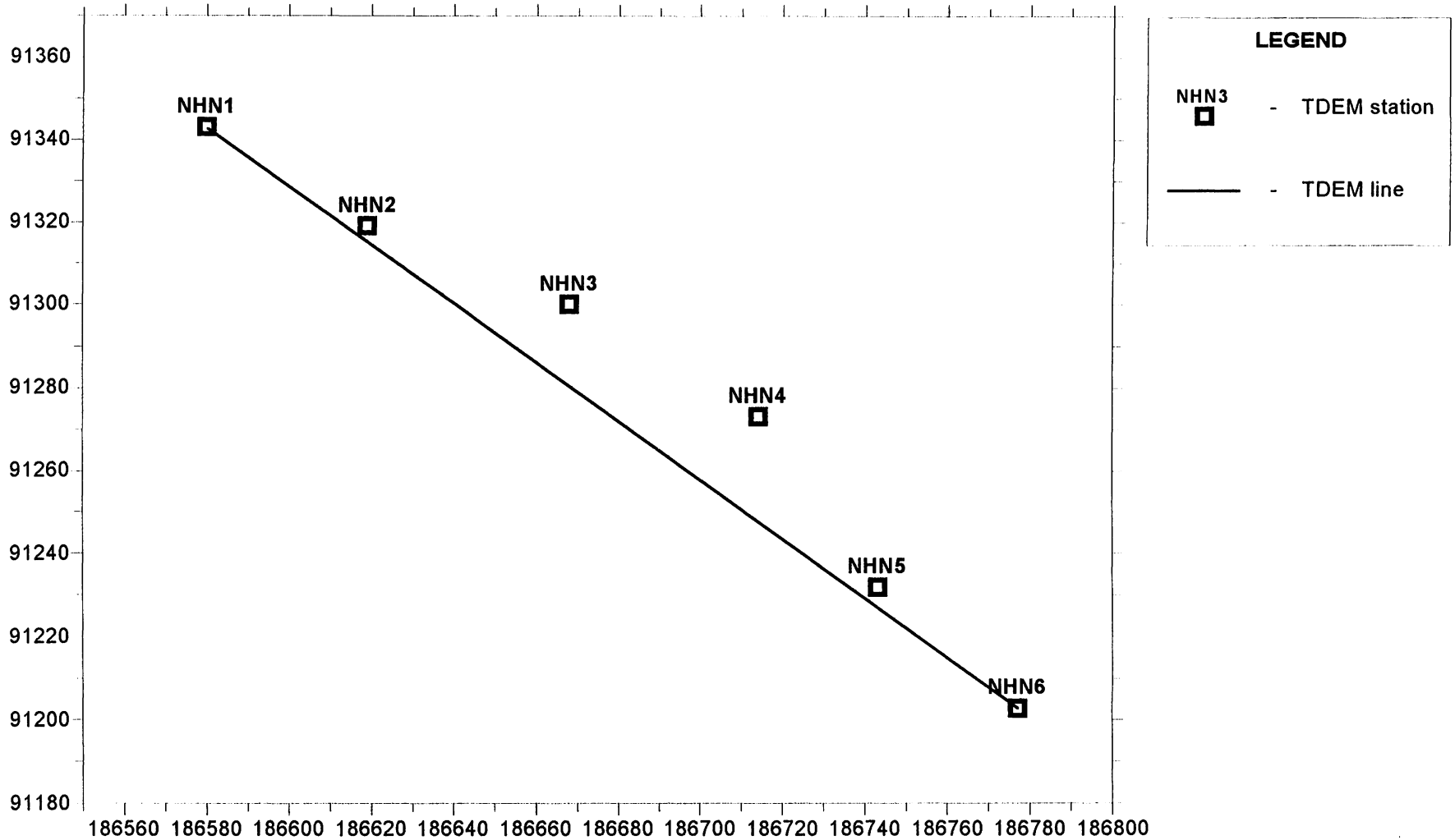


Fig.1 TDEM location map of the Nahal Hever North (NHN) site.

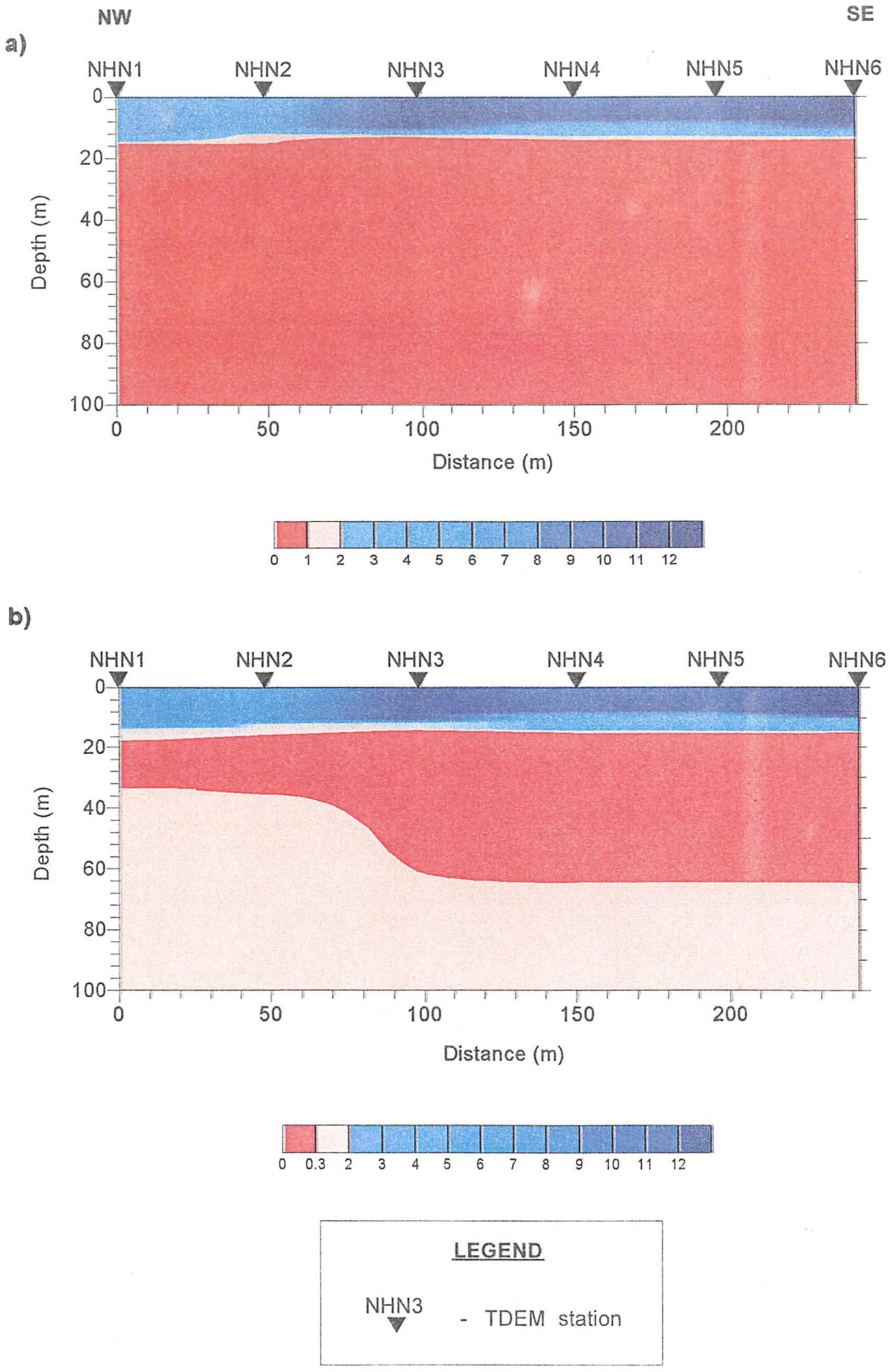


Fig.2 TDEM pseudo-2D resistivity cross-section in the Nahal Hever North (NHN) site (layered inversion).

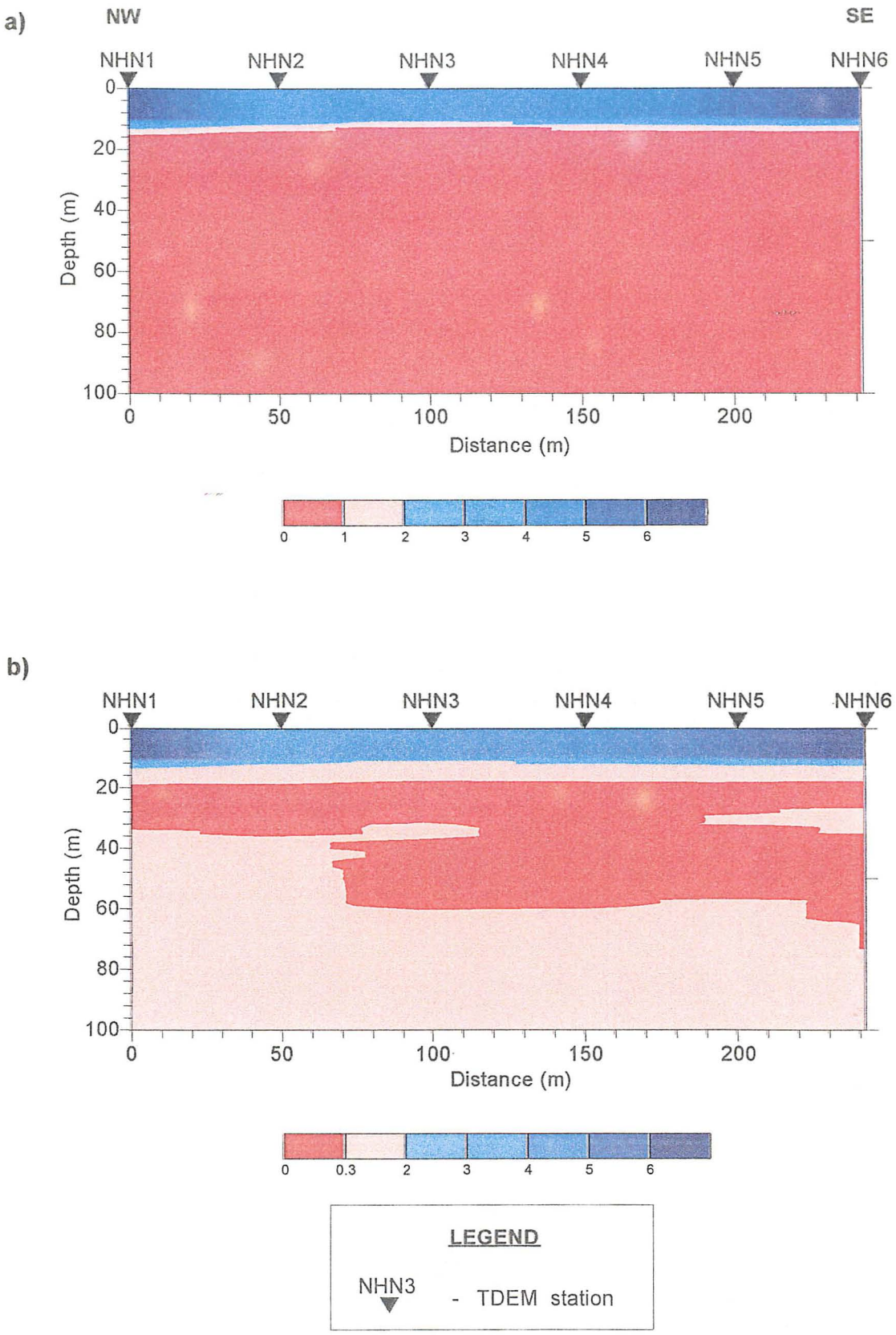


Fig.3 TDEM pseudo-2D resistivity cross-section in the Nahal Hever North (NHN) site (smooth inversion).